地球物理通論第一堂課

CLIL Sentence Bank with Three Examples

附錄二｜CLIL 核心句型

Level 1｜描述現象 Describing Observations

1. We observe that ______.

我們觀察到________。

Example 1 — Plate tectonics

We observe that many earthquakes occur around Taiwan.

我們觀察到台灣周圍發生許多地震。

Example 2 — Topography

We observe that high mountains are concentrated in central Taiwan.

我們觀察到高山集中在台灣中部。

Example 3 — Seismic survey

We observe that the first arrival time increases with distance.

我們觀察到初達時間隨距離增加。

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2. The map shows ______.

這張地圖顯示________。

Example 1

The map shows the topography around Taiwan.

這張地圖顯示台灣周圍的地形。

Example 2

The map shows the distribution of earthquakes.

這張地圖顯示地震的分布。

Example 3

The map shows a deep trench east of Taiwan.

這張地圖顯示台灣東方有一條深海溝。

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3. The data show that ______.

資料顯示________。

Example 1

The data show that earthquakes are not randomly distributed.

資料顯示地震並不是隨機分布。

Example 2

The data show that seismic velocity increases with depth.

資料顯示地震波速度隨深度增加。

Example 3

The data show that gravity varies from place to place.

資料顯示不同地點的重力有所不同。

---

4. There is / There are ______.

有________。

Example 1

There is a deep trench east of the Philippines.

菲律賓東方有一條深海溝。

Example 2

There are many volcanoes around the Pacific Ocean.

太平洋周圍有許多火山。

Example 3

There are several seismic stations along the survey line.

測線上有數個地震儀站。

Level 2｜描述位置 Describing Locations

5. __ is located in ____.

________位於________。

Example 1

Taiwan is located in East Asia.

台灣位於東亞。

Example 2

The Central Mountain Range is located in central Taiwan.

中央山脈位於台灣中部。

Example 3

Iceland is located on the Mid-Atlantic Ridge.

冰島位於中大西洋海脊上。

---

6. __ is located near ____.

________位於________附近。

Example 1

Many earthquakes are located near plate boundaries.

許多地震位於板塊邊界附近。

Example 2

Volcanoes are often located near subduction zones.

火山常位於隱沒帶附近。

Example 3

The seismic stations are located near the survey line.

地震儀位於測線附近。

---

7. __ is located to the east of ____.

________位於________東方。

Example 1

The Philippine Sea Plate is located to the east of Taiwan.

菲律賓海板塊位於台灣東方。

Example 2

The Ryukyu Trench is located to the east of the Ryukyu Islands.

琉球海溝位於琉球群島東方。

Example 3

The Pacific Ocean is located to the east of Taiwan.

太平洋位於台灣東方。

---

8. __ is distributed along ____.

________沿著________分布。

Example 1

Earthquakes are distributed along plate boundaries.

地震沿著板塊邊界分布。

Example 2

Volcanoes are distributed along the island arc.

火山沿著島弧分布。

Example 3

Seismic stations are distributed along the survey line.

地震儀沿著測線分布。

Level 3｜比較 Comparing

9. A is higher than B.

A 比 B 高。

Example 1

The Central Mountain Range is higher than the Western Plain.

中央山脈比西部平原高。

Example 2

The seismic velocity of the lower layer is higher than that of the upper layer.

下層的震波速度比上層高。

Example 3

The temperature near the geothermal area is higher than the surrounding area.

地熱區附近的溫度比周圍高。

---

10. A is deeper than B.

A 比 B 深。

Example 1

The Mariana Trench is deeper than the Taiwan Strait.

馬里亞納海溝比台灣海峽深。

Example 2

Earthquake A is deeper than earthquake B.

地震 A 比地震 B 深。

Example 3

The second layer is deeper than the first layer.

第二層比第一層深。

---

11. A is larger than B.

A 比 B 大。

Example 1

The magnitude of earthquake A is larger than that of earthquake B.

地震 A 的規模比地震 B 大。

Example 2

The gravity anomaly in this region is larger than that in the surrounding area.

這個區域的重力異常比周圍大。

Example 3

The wavelength of the first signal is larger than that of the second signal.

第一個訊號的波長比第二個大。

---

12. Compared with A, B is ______.

與 A 相比，B________ 。

Example 1

Compared with western Taiwan, eastern Taiwan is more mountainous.

與台灣西部相比，東部山地較多。

Example 2

Compared with the upper layer, the lower layer has a higher seismic velocity.

與上層相比，下層具有較高震波速度。

Example 3

Compared with continental crust, oceanic crust is thinner.

與大陸地殼相比，海洋地殼較薄。

Level 4｜因果關係 Cause and Effect

13. __ causes ____.

________造成________ 。

Example 1

Plate movement causes earthquakes.

板塊運動造成地震。

Example 2

Density differences cause gravity anomalies.

密度差異造成重力異常。

Example 3

Temperature differences cause heat flow.

溫度差異造成熱流。

---

14. __ is caused by ____.

________是由________造成的。

Example 1

Many earthquakes are caused by fault movement.

許多地震是由斷層運動造成的。

Example 2

A gravity anomaly can be caused by underground density differences.

重力異常可能由地下密度差異造成。

Example 3

Volcanic activity can be caused by rising magma.

火山活動可能由岩漿上升造成。

---

15. Because __, ____.

因為________，所以________。

Example 1

Because the plates are moving, stress accumulates along faults.

因為板塊正在移動，所以應力會累積在斷層附近。

Example 2

Because the lower layer has a higher velocity, the seismic wave travels faster.

因為下層速度較高，所以地震波傳得更快。

Example 3

Because hot rocks are present underground, the heat flow is high.

因為地下存在高溫岩石，所以熱流較高。

---

16. Therefore, ______.

因此________。

Example 1

The earthquakes form a narrow zone. Therefore, they may be related to a plate boundary.

地震形成狹長帶狀，因此可能與板塊邊界有關。

Example 2

The second layer has a higher velocity. Therefore, refracted waves may arrive earlier at long distances.

第二層速度較高，因此在較遠距離折射波可能先抵達。

Example 3

The gravity value is unusually high. Therefore, dense material may exist underground.

重力值異常偏高，因此地下可能存在高密度物質。

Level 5｜科學證據 Scientific Evidence

17. The evidence shows that ______.

證據顯示________。

Example 1

The evidence shows that the Pacific Plate is moving.

證據顯示太平洋板塊正在移動。

Example 2

The evidence shows that earthquake depth increases toward the west.

證據顯示地震深度往西增加。

Example 3

The evidence shows that seismic velocity changes with depth.

證據顯示震波速度會隨深度改變。

---

18. According to the data, ______.

根據資料________。

Example 1

According to the data, most earthquakes occur near plate boundaries.

根據資料，多數地震發生在板塊邊界附近。

Example 2

According to the data, the eastern part of Taiwan has higher elevations.

根據資料，台灣東部海拔較高。

Example 3

According to the data, the travel time increases with distance.

根據資料，走時隨距離增加。

---

19. This result suggests that ______.

這個結果暗示________。

Example 1

This result suggests that a plate may be subducting beneath another plate.

這個結果暗示某一板塊可能正隱沒到另一板塊之下。

Example 2

This result suggests that a high-density body may exist underground.

這個結果暗示地下可能存在高密度物體。

Example 3

This result suggests that the deeper layer has a higher seismic velocity.

這個結果暗示深層具有較高震波速度。

---

20. This evidence supports the idea that ______.

這項證據支持________的想法。

Example 1

This evidence supports the idea that the seafloor is spreading.

這項證據支持海底正在擴張的想法。

Example 2

This evidence supports the idea that Taiwan is located in an active tectonic region.

這項證據支持台灣位於活躍構造區的想法。

Example 3

This evidence supports the idea that underground structures affect seismic travel times.

這項證據支持地下構造會影響震波走時的想法。

Level 6｜提出假說 Forming Hypotheses

21. One possible explanation is ______.

一個可能的解釋是________。

Example 1

One possible explanation is that two plates are colliding.

一個可能的解釋是兩個板塊正在碰撞。

Example 2

One possible explanation is that dense rocks exist underground.

一個可能的解釋是地下存在高密度岩石。

Example 3

One possible explanation is that the lower layer has a higher seismic velocity.

一個可能的解釋是下層具有較高震波速度。

---

22. We hypothesize that ______.

我們假設________。

Example 1

We hypothesize that earthquakes are related to plate boundaries.

我們假設地震與板塊邊界有關。

Example 2

We hypothesize that the high gravity anomaly is caused by dense rocks.

我們假設高重力異常是由高密度岩石造成。

Example 3

We hypothesize that seismic velocity increases with depth.

我們假設震波速度隨深度增加。

---

23. If our hypothesis is correct, we expect ______.

如果假說正確，我們預期________。

Example 1

If our hypothesis is correct, we expect earthquakes to cluster near the plate boundary.

如果假說正確，我們預期地震會集中在板塊邊界附近。

Example 2

If our hypothesis is correct, we expect higher gravity values above the dense body.

如果假說正確，我們預期高密度物體上方會有較高重力值。

Example 3

If our hypothesis is correct, we expect refracted waves to arrive earlier at long distances.

如果假說正確，我們預期在較遠距離折射波會較早抵達。

Level 7｜測試與驗證 Testing and Verification

24. We can test this idea by ______.

我們可以透過________驗證這個想法。

Example 1

We can test this idea by mapping earthquake locations.

我們可以透過繪製地震位置驗證這個想法。

Example 2

We can test this idea by measuring gravity at different locations.

我們可以透過在不同地點量測重力來驗證這個想法。

Example 3

We can test this idea by comparing seismic travel times.

我們可以透過比較震波走時來驗證這個想法。

---

25. We need more data to determine whether ______.

我們需要更多資料來判斷是否________。

Example 1

We need more data to determine whether the earthquakes define a plate boundary.

我們需要更多資料判斷地震是否界定出板塊邊界。

Example 2

We need more data to determine whether the anomaly is caused by density differences.

我們需要更多資料判斷異常是否由密度差造成。

Example 3

We need more data to determine whether the seismic velocity increases with depth.

我們需要更多資料判斷震波速度是否隨深度增加。

---

26. The result is consistent with our hypothesis.

結果與我們的假說一致。

Example 1

Earthquakes cluster along the boundary. The result is consistent with our hypothesis.

地震集中於邊界附近，結果與我們的假說一致。

Example 2

Gravity increases above the dense body. The result is consistent with our hypothesis.

高密度物體上方的重力增加，結果與假說一致。

Example 3

The refracted wave arrives earlier at long distances. The result is consistent with our hypothesis.

遠距離折射波較早抵達，結果與假說一致。

---

27. The result does not support our hypothesis.

結果不支持我們的假說。

Example 1

The earthquakes are randomly distributed. The result does not support our hypothesis.

地震呈隨機分布，因此結果不支持我們的假說。

Example 2

The measured gravity does not increase. The result does not support our hypothesis.

量測到的重力沒有增加，因此結果不支持假說。

Example 3

The travel times do not match our prediction. The result does not support our hypothesis.

走時與預測不同，因此結果不支持假說。

Level 8｜科學模型 Scientific Modeling

28. Our initial model suggests that ______.

我們最初的模型認為________。

Example 1

Our initial model suggests that Taiwan is located between two interacting plates.

我們最初的模型認為台灣位於兩個交互作用的板塊之間。

Example 2

Our initial model suggests that the subsurface contains two layers.

我們最初的模型認為地下包含兩層。

Example 3

Our initial model suggests that the gravity anomaly is caused by a dense underground body.

我們最初的模型認為重力異常是由地下高密度物體造成。

---

29. Based on the new evidence, we revised our model.

根據新的證據，我們修改了模型。

Example 1

Based on the earthquake-depth data, we revised our model.

根據地震深度資料，我們修改了模型。

Example 2

Based on the seismic travel times, we revised our model.

根據震波走時，我們修改了模型。

Example 3

Based on the new gravity measurements, we revised our model.

根據新的重力量測結果，我們修改了模型。

---

30. The revised model explains ______ better.

修改後的模型更能解釋________。

Example 1

The revised model explains the distribution of earthquakes better.

修改後模型更能解釋地震分布。

Example 2

The revised model explains the travel-time curve better.

修改後模型更能解釋走時曲線。

Example 3

The revised model explains the gravity anomaly better.

修改後模型更能解釋重力異常。

---

31. However, the model cannot explain ______.

但是這個模型無法解釋________。

Example 1

However, the model cannot explain all deep earthquakes.

但是這個模型無法解釋所有深層地震。

Example 2

However, the model cannot explain the gravity anomaly in the northern area.

但是這個模型無法解釋北部區域的重力異常。

Example 3

However, the model cannot explain all observed travel times.

但是這個模型無法解釋所有觀測到的走時。

附錄三｜CER 專用句型

Claim

1. Our claim is that ______.

我們的主張是________。

- Our claim is that earthquake distribution is related to plate boundaries.
- Our claim is that the lower layer has a higher seismic velocity.
- Our claim is that the gravity anomaly is related to underground density.

---

2. We think that ______.

我們認為________。

- We think that Taiwan's mountains are related to plate collision.
- We think that this deep trench is related to subduction.
- We think that the first-arrival pattern reflects underground layering.

---

3. We propose that ______.

我們提出________。

- We propose that the Philippine Sea Plate interacts with the Eurasian Plate near Taiwan.
- We propose that a high-density body exists beneath this area.
- We propose that the subsurface contains at least two seismic layers.

Evidence

4. The evidence shows ______.

證據顯示________。

- The evidence shows a clear earthquake belt east of Taiwan.
- The evidence shows increasing earthquake depth toward the west.
- The evidence shows two different slopes on the travel-time graph.

---

5. We observed ______.

我們觀察到________。

- We observed many earthquakes near the trench.
- We observed higher gravity values in the central area.
- We observed that distant stations received a different first arrival.

---

6. According to the data, ______.

根據資料________。

- According to the data, earthquake depth changes systematically.
- According to the data, the mountain range is concentrated in eastern Taiwan.
- According to the data, seismic velocity is different between the two layers.

Reasoning

7. This evidence supports our claim because ______.

這項證據支持我們的主張，因為________。

- This evidence supports our claim because earthquakes are concentrated near the tectonic boundary.
- This evidence supports our claim because high-density rocks produce stronger gravitational attraction.
- This evidence supports our claim because travel-time slopes are related to seismic velocity.

---

8. If __ is true, we would expect ____.

如果________是真的，我們應該預期________。

- If subduction is occurring, we would expect earthquake depth to increase systematically.

- If a dense body exists underground, we would expect a positive gravity anomaly.
- If the lower layer is faster, we would expect refracted waves to become the first arrivals at longer distances.

---

9. The observed pattern is consistent with ______.

觀察到的型態與________一致。

- The observed pattern is consistent with plate subduction.
- The observed pattern is consistent with a two-layer seismic model.
- The observed pattern is consistent with an underground high-density structure.

CER 完整示例

Example 1｜台灣地震

Claim

Our claim is that earthquake activity around Taiwan is related to plate interactions.

Evidence

The earthquake map shows that earthquakes are concentrated in specific regions around Taiwan.

Reasoning

If earthquakes are related to plate interactions, their locations should follow tectonic structures rather than being randomly distributed.

Example 2｜折射震測

Claim

Our claim is that the lower layer has a higher seismic velocity.

Evidence

The travel-time graph contains two line segments with different slopes.

Reasoning

Because the slope of a travel-time line is related to seismic velocity, the different slopes indicate different seismic velocities.

Example 3｜重力

Claim

Our claim is that a high-density body exists underground.

Evidence

A positive gravity anomaly is observed above the study area.

Reasoning

A high-density body produces greater gravitational attraction, so the positive anomaly supports the claim.

附錄四｜DEAR 科學建模句型

D — Design a Model

1. Our initial model is ______.

我們最初的模型是________。

- Our initial model is a two-layer Earth model.
- Our initial model is a plate-collision model for Taiwan.
- Our initial model is a dense-body model for the gravity anomaly.

---

2. We think the system works like this: ______.

我們認為這個系統可能如此運作：________。

- We think the system works like this: one plate moves beneath another plate.
- We think the system works like this: seismic waves travel through layers with different velocities.
- We think the system works like this: underground density differences change the observed gravity.

E — Elaborate the Model

3. We added ______ to the model.

我們在模型中加入________。

- We added earthquake depth to the model.

- We added a second seismic layer to the model.
- We added a high-density body to the gravity model.

---

4. The model now includes ______.

目前模型包括________。

- The model now includes plate motion and earthquake distribution.
- The model now includes direct waves and refracted waves.
- The model now includes rock density and gravity anomaly.

A — Apply the Model

5. We use the model to predict ______.

我們使用模型預測________。

- We use the model to predict where deep earthquakes may occur.
- We use the model to predict seismic arrival times.
- We use the model to predict the shape of the gravity anomaly.

---

6. According to the model, we expect ______.

根據模型，我們預期________。

- According to the model, we expect deeper earthquakes farther from the trench.
- According to the model, we expect refracted waves to arrive first at long distances.
- According to the model, we expect higher gravity values above denser rocks.

R — Revise the Model

7. The evidence does not fully support our original model.

證據並未完全支持原來的模型。

- The earthquake depths do not fully support our original model.
- The travel times do not fully support our original model.
- The gravity measurements do not fully support our original model.

---

8. Therefore, we revised the model.

因此我們修改了模型。

- Therefore, we revised the plate-boundary model.
- Therefore, we revised the seismic-layer model.
- Therefore, we revised the underground density model.

---

9. Our revised model suggests ______.

修改後的模型認為________。

- Our revised model suggests that the tectonic structure is more complex than we first expected.
- Our revised model suggests that at least three seismic layers may exist.
- Our revised model suggests that more than one density anomaly may exist underground.

附錄五｜AI Prompt 常用句型

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1. Do not give me the final answer directly.

不要直接告訴我最後答案。

Example 1

> Do not give me the final answer directly. Ask me what I observe on the earthquake map first.

Example 2

> Do not give me the final answer directly. Help me identify evidence for plate tectonics.

Example 3

> Do not give me the final answer directly. Guide me through the seismic travel-time problem step by step.

---

2. Ask me questions that can help me think about this problem.

問我一些能幫助我思考這個問題的問題。

Example 1

> Ask me questions that can help me think about why earthquakes occur around Taiwan.

Example 2

> Ask me questions that can help me think about why the travel-time graph has two slopes.

Example 3

> Ask me questions that can help me think about the source of this gravity anomaly.

---

3. Suggest three possible explanations.

提出三個可能的解釋。

Example 1

> Suggest three possible explanations for the concentration of earthquakes east of Taiwan.

Example 2

> Suggest three possible explanations for this positive gravity anomaly.

Example 3

> Suggest three possible explanations for the high heat flow in this region.

---

4. What evidence would support this explanation?

什麼證據可以支持這個解釋？

Example 1

> What evidence would support the explanation that a plate is subducting?

Example 2

> What evidence would support the explanation that dense rocks exist underground?

Example 3

> What evidence would support the explanation that the lower layer has a higher seismic velocity?

---

5. What evidence would contradict this explanation?

什麼證據會與這個解釋矛盾？

Example 1

What evidence would contradict the subduction model?

Example 2

What evidence would contradict the two-layer seismic model?

Example 3

What evidence would contradict the hypothesis of an underground high-density body?

---

6. Check my reasoning. Do not rewrite my answer.

檢查我的推理，但不要直接幫我重寫答案。

Example 1

Check my reasoning about plate tectonics. Do not rewrite my answer.

Example 2

Check my reasoning about seismic refraction. Do not rewrite my answer.

Example 3

Check my reasoning about the gravity anomaly. Do not rewrite my answer.

---

7. What is the weakest part of my explanation?

我的解釋中最薄弱的地方是什麼？

Example 1

What is the weakest part of my explanation for Taiwan's frequent earthquakes?

Example 2

What is the weakest part of my interpretation of this seismic profile?

Example 3

What is the weakest part of my explanation for this geothermal anomaly?

---

8. If this model is correct, what should we observe?

如果這個模型正確，我們應該觀察到什麼？

Example 1

If the subduction model is correct, what should we observe?

Example 2

If the two-layer seismic model is correct, what should we observe?

Example 3

If the dense-body gravity model is correct, what should we observe?

附錄六｜小組討論常用英文

1. What do you think?

你怎麼想？

- What do you think about this earthquake distribution?
- What do you think this color represents?
- What do you think about the AI explanation?

---

2. Why do you think so?

你為什麼這樣想？

- Why do you think this is a plate boundary?
- Why do you think the lower layer is faster?
- Why do you think this gravity anomaly is important?

---

3. What is your evidence?

你的證據是什麼？

- What is your evidence for plate collision?
- What is your evidence for a two-layer structure?
- What is your evidence for an underground density anomaly?

---

4. I agree because ______.

我同意，因為________。

- I agree because the earthquake data support your idea.
- I agree because the map shows a clear pattern.

- I agree because the travel-time graph matches the model.

---

5. I disagree because ______.

我不同意，因為________。

- I disagree because the earthquakes are not located near the proposed boundary.
- I disagree because the data do not show a clear trend.
- I disagree because our measurement does not match the prediction.

---

6. Can you explain that again?

你可以再解釋一次嗎？

- Can you explain the plate-motion model again?
- Can you explain the travel-time graph again?
- Can you explain how the gravity anomaly is produced again?

---

7. Let's compare the two ideas.

我們比較這兩個想法。

- Let's compare the collision model and the subduction model.
- Let's compare our model with the AI-generated model.
- Let's compare the direct-wave model with the refracted-wave model.

---

8. Let's ask AI, but we need to verify the answer.

我們可以問 AI，但需要驗證答案。

- Let's ask AI about this earthquake pattern, but we need to verify the answer.
- Let's ask AI to explain the code, but we need to verify the answer.
- Let's ask AI about the gravity anomaly, but we need to verify the answer.

---

9. Let's check the data.

我們檢查資料。

- Let's check the earthquake-depth data.
- Let's check the PyGMT map.
- Let's check the seismic travel-time data.

---

10. I found a different result.

我得到不同結果。

- I found a different earthquake distribution.
- I found a different travel time.
- I found a different value in the gravity data.

---

11. Which explanation is better supported by the evidence?

哪一個解釋得到較多證據支持？

- Which explanation of Taiwan's earthquakes is better supported by the evidence?
- Which underground model is better supported by the travel-time data?
- Which density model is better supported by the gravity measurements?

附錄七｜教師課堂管理英文

1. Please work in groups of two or three.

請兩到三人一組。

- Please work in groups of two or three for this activity.
- Please work in groups of two or three and compare your maps.
- Please work in groups of two or three and discuss your hypothesis.

---

2. Choose one group leader.

選一位組長。

- Choose one group leader for today's activity.
- Choose one group leader to organize the files.
- Choose one group leader to upload your final work.

---

3. Please open your laptop.

請打開筆電。

- Please open your laptop and log in to Google Colab.
- Please open your laptop and go to GitHub.
- Please open your laptop and start Antigravity.

---

4. Please follow my demonstration.

請跟著我的示範。

- Please follow my demonstration and create a new notebook.
- Please follow my demonstration and run the PyGMT code.
- Please follow my demonstration and create a repository.

---

5. You have ten minutes to complete this task.

你們有十分鐘完成這項任務。

- You have ten minutes to complete the GitHub task.
- You have ten minutes to discuss your hypothesis.
- You have ten minutes to complete the CER activity.

---

6. Stop here for a moment.

先做到這裡。

- Stop here for a moment and look at the screen.
- Stop here for a moment. I want to explain this result.
- Stop here for a moment and compare your answer with your group members.

---

7. Don't worry if you make a mistake.

犯錯沒有關係。

- Don't worry if you make a mistake in the code.
- Don't worry if your first model is incorrect.
- Don't worry if your result is different from mine.

---

8. Please show me your screen.

請讓我看你的畫面。

- Please show me your screen if the code does not work.
- Please show me your screen when your map is finished.
- Please show me your screen if you cannot find the error.

---

9. Raise your hand if you need help.

需要協助請舉手。

- Raise your hand if you need help with GitHub.
- Raise your hand if you need help with the Python code.
- Raise your hand if you need help understanding the question.

---

10. Please save your work.

請儲存你的工作。

- Please save your Colab notebook.
- Please save your PyGMT figure.
- Please save your AI conversation.

---

11. Please keep your AI conversation.

請保留 AI 對話紀錄。

- Please keep your AI conversation as part of your assignment.
- Please keep your AI conversation so we can review your reasoning process.
- Please keep your AI conversation and record how your ideas changed.

---

12. Please upload your work before you leave.

離開前請上傳你的作業。

- Please upload your notebook before you leave.
- Please upload your AI interaction log before you leave.
- Please upload your group results before you leave.

附錄八｜整學期反覆使用的10個核心句型

這十句建議固定出現在：

- PPT
- 學習單
- Google Colab
- GitHub Pages
- 作業評量規準
- 小組報告

讓學生真正把語言變成「科學思考工具」。

---

1. We observe that ______.

我們觀察到________。

- We observe that earthquakes are concentrated around Taiwan.
- We observe that earthquake depth increases toward the west.
- We observe that the travel time increases with distance.

---

2. The data show that ______.

資料顯示________。

- The data show that earthquakes are not randomly distributed.
- The data show that seismic velocity changes with depth.
- The data show that gravity varies across the study area.

---

3. One possible explanation is ______.

一個可能的解釋是________。

- One possible explanation is plate collision.
- One possible explanation is seismic refraction.
- One possible explanation is an underground density difference.

---

4. We hypothesize that ______.

我們假設________。

- We hypothesize that Taiwan's earthquakes are related to plate interactions.
- We hypothesize that the lower layer has a higher seismic velocity.
- We hypothesize that dense rocks exist beneath the study area.

---

5. If our hypothesis is correct, we expect ______.

如果假說正確，我們預期________。

- If our hypothesis is correct, we expect earthquakes to cluster near tectonic boundaries.
- If our hypothesis is correct, we expect refracted waves to arrive first at long distances.
- If our hypothesis is correct, we expect a positive gravity anomaly above the dense body.

---

6. The evidence supports our hypothesis because ______.

證據支持我們的假說，因為________。

- The evidence supports our hypothesis because the earthquakes form a clear belt.
- The evidence supports our hypothesis because the observed travel times match the model.
- The evidence supports our hypothesis because high gravity values occur above the proposed dense body.

---

7. However, ______ cannot be explained by our model.

但是________無法由目前模型解釋。

- However, several deep earthquakes cannot be explained by our model.
- However, some travel times cannot be explained by our model.
- However, the northern gravity anomaly cannot be explained by our model.

---

8. Therefore, we revised our model.

因此我們修改模型。

- Therefore, we revised our plate-tectonic model.
- Therefore, we revised our seismic-layer model.
- Therefore, we revised our underground density model.

---

9. We need more data to determine whether ______.

我們需要更多資料確認是否________。

- We need more data to determine whether this line represents a plate boundary.
- We need more data to determine whether a third seismic layer exists.
- We need more data to determine whether the anomaly is caused by density differences.

---

10. AI suggested __, but we verified it by ____.

AI 建議________，但是我們透過________進行驗證。

Example 1 — Plate tectonics

> AI suggested that the earthquake pattern is related to plate subduction, but we verified it by comparing earthquake depth with the plate boundary.

AI 建議地震型態與板塊隱沒有關，但我們透過比較地震深度與板塊邊界進行驗證。

Example 2 — PyGMT

> AI suggested that my PyGMT code was incorrect, but we verified it by checking the official documentation and running the revised code.

AI 建議我的 PyGMT 程式有錯，但我們透過查閱官方文件並重新執行程式進行驗證。

Example 3 — Seismic refraction

> AI suggested that the second arrival was a refracted wave, but we verified it by comparing the observed travel time with the theoretical prediction.

AI 建議第二個訊號是折射波，但我們透過比較實際走時與理論預測進行驗證。

建議學生最後熟練的「科學論述六句組」

如果學生英文程度有限，不需要一開始背全部句型。

先讓學生固定使用下面六句，就已經可以完成一個基本科學探究報告。

① Observation

> We observe that ______.

我們觀察到________。

② Question

> We want to know why ______.

我們想知道為什麼________。

③ Hypothesis

> We hypothesize that ______.

我們假設________。

④ Prediction

> If our hypothesis is correct, we expect ______.

如果假說正確，我們預期________。

⑤ Evidence

The evidence shows that ______.

證據顯示________。

⑥ Conclusion / Revision

Therefore, we support / revise our model.

因此我們支持／修改我們的模型。

六句完整範例｜板塊構造

We observe that many earthquakes occur around Taiwan.

We want to know why earthquakes are concentrated in these regions.

We hypothesize that the earthquake distribution is related to plate interactions.

If our hypothesis is correct, we expect earthquakes to follow tectonic structures.

The evidence shows that earthquake locations form several clear belts.

Therefore, we support and further revise our model of plate interactions around Taiwan.

六句完整範例｜折射震測

We observe that travel time increases with distance.

We want to know why the travel-time graph has two different slopes.

We hypothesize that the subsurface contains two layers with different seismic velocities.

If our hypothesis is correct, we expect the refracted wave to become the first arrival at longer distances.

The evidence shows that the distant stations follow a second travel-time line.

Therefore, we support the two-layer model.

六句完整範例｜重力探勘

We observe that gravity is higher in the central part of the study area.

We want to know why this positive gravity anomaly occurs.

We hypothesize that a high-density body exists underground.

If our hypothesis is correct, we expect the highest gravity values above the dense body.

The evidence shows that the observed gravity anomaly matches the predicted pattern.

Therefore, we support the high-density-body model.

最核心的三句課堂口號

Science

How do we know?

我們怎麼知道？

Examples

How do we know that plates are moving?

How do we know the structure beneath the ground?

How do we know that the AI answer is correct?

Evidence

What evidence supports this idea?

什麼證據支持這個想法？

Examples

What evidence supports plate tectonics?

What evidence supports the two-layer seismic model?

What evidence supports your interpretation?

AI Literacy

> **AI can generate an answer, but understanding the answer is your responsibility.**

AI 可以產生答案，但是理解答案是你的責任。

Examples

> AI can generate a map, but you must understand what the map shows.
>
> AI can write the code, but you must verify whether the code is correct.
>
> AI can suggest an explanation, but you must decide whether the evidence supports it.